

1.Program 1: OOP Basics - Class, Object, Encapsulation, Constructor

```
9
10
11 class Student{
12     private String name;
13     private int age;
14
15     public Student(String name,int age){
16         this.name=name; //1
17         this.age=age;
18     }
19     public String getName(){
20         return name;
21     }
22     public void setAge(int age){
23         this.age=age; //2
24     }
25     public void display(){
26         System.out.println("Name: "+name+"Age: "+age); //3
27     }
28 }
29
```

OUTPUT

```
Name: ArunAge: 21
```

2. Program 2: Method Overloading

```
1 public class main{
2     public static void main(String[] args){
3         Calculator calc=new Calculator();
4         System.out.println(calc.multiply(2,3)); //2
5         float result = calc.multiply(2.3f,3.5f); //3
6         System.out.println(result);
7     }
8 }
9
10 class Calculator{
11     int multiply(int a,int b){
12         return a*b;
13     }
14     int multiply(int a,int b,int c){
15         return a*b*c; //1
16     }
17     long multiply(long a,long b){
18         return a*b;
19     }
20     float multiply(float a,float b){
21         return a*b;
22     }
23 }
```

OUTPUT

```
6
8.05
```

3.Program 3: Method Overriding

```
12     return "Animal makes a sound";
13 }
14 public void eat(String food){
15     System.out.println("Animal eats "+food);
16 }
17 public String getInfo(){
18     return "Generic animal";
19 }
20 }
21
22 class Dog extends Animal{
23     public String makeSound(){ //1
24         return "Dog Barks";
25     }
26     public void eat(String food){ //2
27         System.out.println("Dog eats "+food);
28     }
29     public String getInfo(){ //3
30         return "Dog info";
31     }
32 }
```

OUTPUT

```
Dog Barks
Dog eats bone
Dog info
```